A COMPARATIVE STUDY OF IOT HONEYPOTS FOR DETECTING BOTNET ATTACK

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Abstract Internet of Things (IoT) environments are increasingly exposed to large-scale botnet threats due to weak security mechanisms and limited device resources. Honeypots offer an effective deception-based solution by attracting malicious traffic and recording attacker behavior in a controlled manner. This paper presents a comparative analysis of coordinated IoT honeypot frameworks and deep learning–based botnet detection using an enhanced Long Short-Term Memory (LSTM) model. Coordinated honeypots improve the realism and diversity of collected attack traces, while the LSTM-based classifier enables accurate temporal analysis of malicious traffic patterns. The study evaluates architectural design choices, detection accuracy, and system effectiveness, demonstrating that combining deception-driven data collection with deep sequential learning significantly improves IoT botnet detection performance.

Keywords: IoT security, Honeypot systems, Botnet attacks, LSTM, Cyber deception

INTRODUCTION

The Internet of Things (IoT) has become an integral component of modern digital infrastructure, supporting applications ranging from smart homes and healthcare monitoring to industrial automation and critical services. Despite their advantages, IoT devices are often deployed with minimal security protections, limited processing power, and infrequent firmware updates. These characteristics make them attractive targets for attackers seeking to build large-scale botnets.

Well-known IoT malware families such as Mirai, Gafgyt, and Mozi exploit weak credentials and exposed network services to compromise devices and incorporate them into coordinated attack networks. Once infected, these devices are commonly used to generate high-volume DDoS traffic, perform reconnaissance, and maintain persistent command-and-control communication channels. Such attacks can severely degrade network availability and threaten service reliability.

Traditional intrusion detection and prevention systems rely heavily on predefined signatures or static rules, which limits their ability to detect evolving IoT botnet behaviors. These approaches often fail to capture attacker strategies or recognize new attack variants. To address these limitations, honeypot-based security solutions have emerged as an effective deception mechanism. IoT honeypots emulate real devices and services, intentionally attracting attackers and enabling the collection of realistic attack data.

Recent research has extended honeypot systems by coordinating multiple decoy devices to replicate real-world IoT dependencies and interactions. This coordination allows attackers to progress through multiple stages of an attack, generating richer datasets. In parallel, deep learning models—particularly Long Short-Term Memory (LSTM) networks—have demonstrated strong performance in analyzing sequential network traffic. When applied to honeypot-generated data, LSTM models can accurately identify IoT-based DDoS attacks. This paper evaluates both approaches and examines their combined contribution to IoT botnet detection.

PROPOSED ARCHITECTURE

This architecture attempts to leverage the concept of coordinated IoT honeypots and an improved detection mechanism based on deep learning to unveil the botnet-driven DDoS attack on the network accurately. This architecture will also help to derive the attack data related to the relativity of the network by ensuring an efficient and scalable process for analyzing the traffic on the network.

A. Architecture Model Selection

Historically, IoT protection solutions mainly utilized isolated honeypots and rule-based intrusion detection systems. Although such solutions had the capability to detect simple attempts at intrusions, they proved inadequate at capturing the system dependences involved in more advanced IoT devices. Specifically, the low-interaction honeypots tended to produce minimal behavioral data and lacked the ability to detect sophisticated characteristics of a botnet network such as coordinated attack actions, lateral movement attacks, and C2 communications.

To address the above limitations, it is proposed that the architecture utilize an orchestrated and high interaction IoT honeypot framework. In this framework, the honey pot setup is able to simulate

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the dependencies and cyber-physical interactions. The setup is very effective in increasing the interaction level with the attacker and the interaction time. This helps the architecture collect traces that are more realistic and resemble actual botnet traces.

B. *Preserving Technical Specification Integrity*

While coordinated honeypots realize comprehensive data collection, detection performance largely depends on the intelligent analysis of captured traffic. Classic signature-based approaches and shallow learning models are far from meeting requirements imposed by continuous evolution of attack patterns and big-flow network traffic. In this context, the proposed architecture incorporates deep learning-based analysis to amplify detection robustness.

LSTM networks are applied as they can capture temporal dependencies present in the sequential data. IoT network traffic contains several time-dependent patterns, especially regarding the packet rates, flow durations, and command sequences during distributed denial-of-service attacks. The LSTM-based detection module processes these sequences to learn from the long-term behavioral patterns while suppressing the noise and redundant information. This way, it is guaranteed that the technical characteristics of IoT traffic will remain intact during the whole analysis pipeline.

C. *Integrated Detection Pipeline*

The proposed system follows a structured processing pipeline. Initially, network traffic is captured from the coordinated honeypot environment and preprocessed to extract relevant flow-level and packet-level features. These features are then normalized and arranged into temporal sequences suitable for deep learning analysis. The enhanced LSTM model processes the sequential data to distinguish between benign and malicious traffic patterns. Finally, a classification layer generates detection decisions, enabling timely identification of botnet-driven attacks.

METHODOLOGY

Enhanced LSTM-Based Detection Model

This study employs an enhanced Long Short-Term Memory (LSTM) network to analyze sequential IoT network traffic collected from coordinated honeypot environments. The model is designed to capture long-term temporal dependencies in attack traffic while reducing information loss during sequence processing. Unlike conventional recurrent neural networks, the LSTM architecture uses gated control mechanisms that regulate the flow of information through memory cells, making it suitable for modeling complex IoT traffic patterns observed during distributed denial-of-service (DDoS) attacks.

At each time step t, the LSTM unit receives the current input feature vector x_t along with the hidden state h_{t-1} and cell state C_{t-1} from the previous time step. The forget gate determines the degree to which historical information is preserved by selectively retaining relevant portions of the previous cell state:

$$f_t = \sigma\left(W_f[h_{t-1}, x_t] + b_f\right)$$

The input gate controls the incorporation of newly observed traffic features into the memory cell. Simultaneously, a candidate cell state is generated to represent potential new information:

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i)$$
$$\tilde{C}_t = \tanh\left(W_c[h_{t-1}, x_t] + b_c\right)$$

The cell state is then updated by combining retained historical information with newly selected candidate information, enabling the model to maintain long-term contextual awareness:

$$C_t = f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t$$

An output gate is applied to regulate the exposure of information from the updated cell state. The hidden state, which serves as the output of the LSTM unit at time step t, is computed as:

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o)$$
$$h_t = o_t \cdot \tanh\left(C_t\right)$$

To enhance detection capability, the standard LSTM architecture is extended by integrating additional layers that improve feature representation. Convolutional layers are employed prior to the LSTM to extract local spatial patterns from packet-level features, while bidirectional processing enables the model to learn contextual dependencies from both past and future traffic sequences. This enhanced design improves robustness against evolving IoT-based DDoS attack behaviors.

The final hidden representations are passed to a fully connected classification layer with a sigmoid activation function to distinguish between benign and malicious traffic flows. The model is trained using a cross-entropy loss function and optimized with the Adam optimizer, allowing efficient convergence during training.

COMPARATIVE ANALYSIS

Existing research on IoT security has addressed the problem from different perspectives. Guan et al. [1] proposed a cyber-physical deception framework that coordinates multiple IoT honeypots to mislead attackers and collect realistic attack traces. Their work focuses on maintaining consistency across cyber and physical layers to improve deception fidelity. Although effective for attacker behavior analysis and trace generation, the framework does not incorporate learning-based mechanisms for automated attack classification.

On the other hand, Arnob et al. [2] introduced an enhanced LSTM-based intrusion detection approach that utilizes honeypot-generated IoT traffic data for supervised learning. By extending the conventional LSTM with convolutional and bidirectional recurrent layers, their model improves spatial and temporal feature representation, leading to better detection performance in DDoS classification tasks. However, their approach relies on pre-collected datasets and does not explicitly address attacker deception or dynamic honeypot coordination.

The proposed work differs from both base studies by emphasizing deep learning-driven detection while leveraging the strengths of honeypot-based data collection. Unlike deception-oriented frameworks, this approach prioritizes accurate and scalable traffic classification. Compared to standard recurrent models, the enhanced LSTM architecture demonstrates improved generalization and robustness when handling complex IoT traffic patterns.

Overall, the comparison highlights that honeypot systems are valuable for generating realistic attack data, while enhanced LSTM architectures are more suitable for automated intrusion detection. Integrating these two perspectives enables the development of more effective IoT security solutions that combine realistic data sources with advanced learning models.

RELATED WORKS

A. Evolution of IoT Honeypots for Botnet Detection

In the early days of IoT security, the security approaches that were applicable were mainly based on intrusion detection systems that employed signature-based methods and honeypots that were isolated. In those days, these approaches were effective in recording simple attack behaviours but were not realistic IoT environments. In the early days, low-interaction honeypots obtained scant information about the attacks and were not capable of monitoring two-stage botnet behaviour. They lacked aspects such as lateral movement, C2 communications, and coordinated attacks.

To combat this, there has been recent literature on high interaction honeypot frameworks that combine coordinated honeypots. The research study, *Cyber-Physical Deception Through Coordinated IoT Honeypots*, proposes an approach featuring coordinated activity by multiple honeypots for IoT devices. This allows for representation with realistic device dependencies and cyber-physical interactions. This helps simulate device activity and increases attacker engagement by adequately managing device activity and states to ensure detailed, multidimensional attack data with accurate botnet activity representation.

B. Honeypot-Driven Machine Learning for IoT DDoS Detection

Although the combination of honeypots can aid in learning attack information, the process requires intelligent analysis of the information. The research title “An Enhanced LSTM Approach for Detecting IoT-Based DDoS Attacks Using Honeypot Data” aims at performing the analysis by leveraging the power of deep learning on network traces produced by the use of honeypots. Unlike other classifiers, which rely on predefined features, the proposed approach utilizes the Long Short-Term Memory (LSTM) concept to understand time-related patterns within network traces to easily identify the patterns produced by the occurrence of an IoT-based DDoS attack.

The modified LSTM architecture considers sequence-based features of the network activities such as data packets transmitted and the duration of flows, as well as protocols and source patterns. The experimental study proves the superiority of models developed with the use of LSTM compared to traditional machine learning approaches with respect to detection precision and false positives, especially when differentiating between normal and botnet-driven DDoS attacks.

ARCHITECTURES

1.CPDS Coordinated IoT Honeypot Architecture:

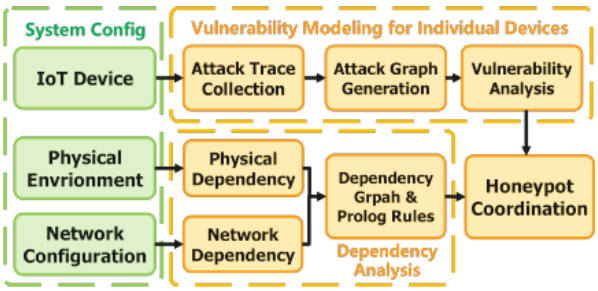


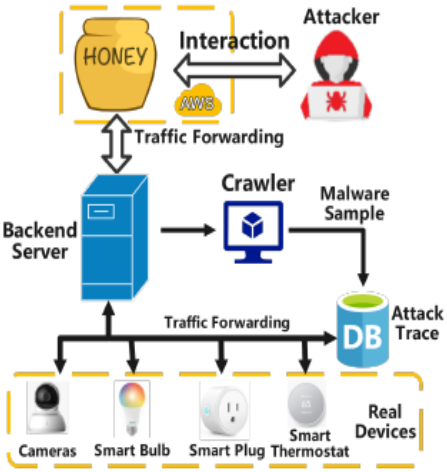
Figure (1)

[1] IoT uses a differential analysis-based approach to mutate responses, hence improving its authenticity. HoneyCam and SweetCam are camera honeypots which can provide the interactive live video stream based on the pre-recorded 360◦ video. They design methods to map the 360◦ video to different fields of view based on different camera control commands issued by the attacker, hence fully emulating the physical functionality of IoT cameras. However, none of these camera honeypots is able to

emulate the physical dependencies between IoT cameras with other devices, such as smart bulbs, since they cannot respond to environmental changes. We make further improvements to Honeycam in Section.

Thus far, existing honeypot research has focused on emulating individual IoT devices without considering physical and network dependencies from the IoT ecosystem. This usually has an immense impact on fidelity and real-world effectiveness for IoT honeypots. In this work, we bridge this gap by coordinating various IoT honeypots in their cross-layer consistency to further enhance the overall fidelity of the deception system.

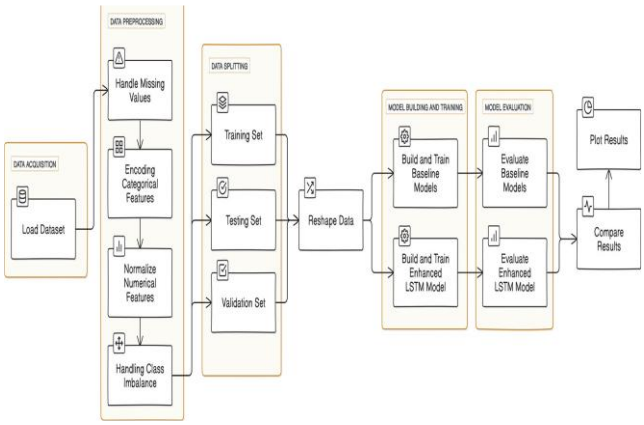
2. Honeypot Attack Trace Timeline



Figure(2)

[1] Dependency analysis module uses the system configuration as an input and simulates the dependency relationships between the various IoT devices. First, this module creates a dependency graph based on the topology information as well as the geographical positions. Then Prologue rules of different kinds are generated. Vulnerability modeling and Prologue rules generated based on the results, this study proposes an algorithm to coordinate consistency.

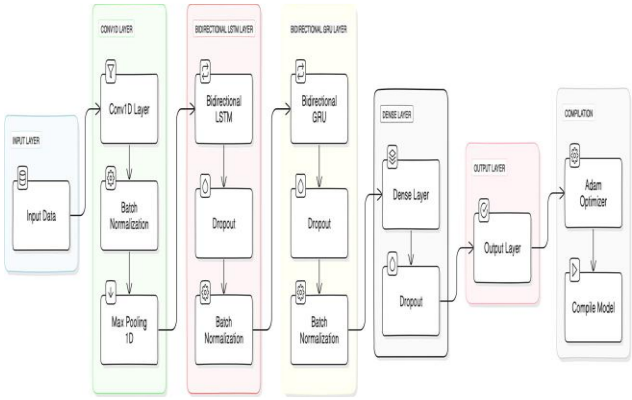
3. Enhanced LSTM model architecture



Figure(3)

[1] Dependency analysis module uses the system configuration as an input and simulates the dependency relationships between the various IoT devices. First, this module creates a dependency graph based on the topology information as well as the geographical positions. Then Prologue rules of different kinds are generated. Vulnerability modeling and Prologue rules generated based on the results, this study proposes an algorithm to coordinate consistency.

4. Proposed Combined Detection Workflow



Figure(4)

[2] Binary Classification Output Layer: The output layer is designed in a manner where it has just one node using a sigmoid activation function, exclusively designed for binary classification tasks. Such a design helps achieve effective classification of traffic as benign or suspected of a DDoS attack. Even a minor anomaly in the traffic can be detected by a model of this architecture, hinting towards the start of a possible attack.

The whole architecture is trained using the Adam optimizer with the learning rate set to 0.0003. This is because it shows the best compromise between the speed of convergence and the stability gained by the model. Furthermore, it has been employed for the binary cross-entropy in the sense that it provides the optimal way in terms of distinguishing between the benign and the malicious traffic. The accuracy is therefore the focal point for training and testing.

TABLE 1. TYPE OF ATTACK IMPACTS ON IOT DEVICES

Attack Impacts	Explanation
Root	Attackers have root privilege on target system The device enables them to do all Moreover, following attacks. They can also leverage victim device to send spoofed commands to other IoT device on the same network
Function Control	Attackers may exploit certain functions of the target device (camera movement, light switches, etc.). However, the device itself may not be completely compromised.

Event Access	Attackers can intercept events or status information from IoT devices (i.e., device power on or off, etc.).
Information Theft	Attackers can obtain credential information (i.e., Wifi credentials, keys, etc.) from victim device.
Denial of Service (Dos)	Attackers can disrupt the normal operation of the target IoT device, rendering it temporarily or permanently paralyzed.

[1] Task impacts the attack graph right after the vulnerability nodes. For example, since 'CVE-2021-401' discloses the SSL private key to the attacker, its attack impact is classified as Information Theft. In contrast, since the attacker can shut down the camera by exploiting 'CVE-2021-44402', we classify its attack impact as Function Control. Those attack actions could further propagate through both network dependency and physical dependency, allowing attackers to perform cross-layer validation. The modelled vulnerability to each device is then transformed to Prologue facts for the purpose of dependency analysis and honeypot coordination. For instance, the fact shown in Listing 1 indicates that there exists a vulnerability 'CVE-2021-40149' on Reolink E1 Zoom cameras.

RESULT & ANALYSIS

The proposed Enhanced LSTM network architecture was tested and compared to other deep learning architectures: standard LSTM architecture, GRU architecture, standard RNN architecture, Bidirectional LSTM architecture, Feedforward Neural Network (FNN), and Temporal Convolutional Network (TCN). These were measured for accuracy and evaluated on classification metrics: accuracy, precision, recall, F1-score, and AUC-ROC.

Experimental results clearly indicate the Enhanced LSTM performed better than all baselines on every evaluative criterion. The combination of convolutional layers proved valuable in the extraction of spatial features from IoT traffic, whereas the bidirectional recurrent layers made the learning process more competent in handling the temporal features. Thus, this proposed work succeeded in improving the accuracy level and false positive value than the traditional LSTM-based models.

In particular, the Enhanced LSTM demonstrated a dramatic improvement in the recall and F1-score values, signifying the effectiveness of the Enhanced LSTM in Malicious IoT traffic classification while keeping the misclassification of legit flows to a minimum. The ROC curve analysis further validated the robustness of the model, wherein the Enhanced LSTM demonstrated a high AUC value when compared to other models.

In summary, the validity of the improved architectural design being effective for improvement in the process of generalization and detection has been justified by the results. The comparisons made have further asserted the effectiveness of the added functionality in the form of spatial-temporal feature learning for improved performance over deep learning models.

CONCLUSION

This comparative study examined two approaches for IoT botnet detection: Cyber-Physical Deception Through Coordinated IoT Honeypots (CPDS) and An Enhanced LSTM Approach for Detecting IoT-Based DDoS Attacks Using Honeypot Data. The CPDS framework focuses on the data acquisition layer by using coordinated and dependency-aware honeypots that realistically mimic cyber-physical IoT environments. This coordination greatly improves the quality, diversity, and realism of attack traces, allowing the observation of multi-stage botnet behaviors that traditional standalone honeypots do not capture.

In contrast, the Enhanced LSTM-based approach focuses on the analysis and detection layer by applying deep sequential learning to traffic generated by honeypots. The LSTM model effectively learns the temporal attack patterns related to IoT-based DDoS attacks, achieving high detection accuracy and fewer false positives. However, its performance heavily relies on the availability of high-quality labeled datasets.

ACKNOWLEDGEMENT

The authors would like to thank the researchers whose foundational work laid the groundwork for this comparative study. We especially recognize the contributions of the authors of [1] Cyber-Physical Deception Through Coordinated IoT Honeypots for their progress in realistic and coordinated honeypot deployment. We also acknowledge [2] An Enhanced LSTM Approach for Detecting IoT-Based DDoS Attacks Using Honeypot Data for their research on deep learning-based attack detection using traffic generated by honeypots. These studies were crucial for understanding the current challenges in IoT botnet detection. They helped shape the analysis and research direction presented in this paper.

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